

BGP Message Types:

BGP basically uses four message types in its operations:

1. Open:

- Open message is sent after Border Gateway Protocol neighbor is configured.
- Exchanges Dynamic Border Gateway Protocol (BGP) values and capabilities.
- Open message is sent to establish or form peering with that BGP neighbor.
- Contains information such as Version, AS Number, Router ID and the Hold-Time.

2. Update:

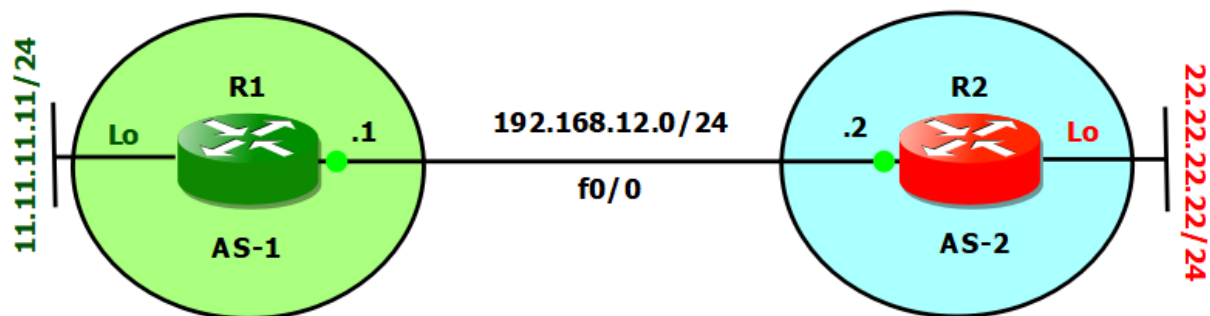
- Routing information between peers is transferred using the Update message.
- Update message include, new routes, withdrawn routes and the path attributes.

3. Keepalive:

- Keepalive message is similar to Hellos message in other routing protocols.
- BGP uses keepalives message which help in keeping peering session active.
- Keepalive messages are sent periodically to maintain neighbor relationship.
- The BGP peers exchange Keepalive messages after every 60 seconds.
- The Border Gateway Protocol Peers Hold Timer is 180 seconds by default.

4. Notification:

- In event of a problem which causes the router to end the BGP peering session.
- The BGP neighbor send a notification message and the connection is closed.
- Border Gateway Protocol use notification to resets the neighbor relationship.
- Such as Bad peer AS Number, bad BGP Identifier, unacceptable hold time etc.



R1 Basic Configuration

```
R1(config)#interface f0/0
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface loopback 1
R1(config-if)#ip add 11.11.11.11 255.255.255.0
```

R2 Basic Configuration

```
R2(config)#interface f0/0
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface loopback 2
R2(config-if)#ip add 22.22.22.22 255.255.255.0
```

R1 Configuration

```
R1(config)#router bgp 1
R1(config-router)#neighbor 192.168.12.2 remote-as 2
R1(config-router)#network 11.11.11.0 mask 255.255.255.0
```

R2 Configuration

```
R2(config)#router bgp 2
R2(config-router)#neighbor 192.168.12.1 remote-as 1
R2(config-router)#network 22.22.22.0 mask 255.255.255.0
```

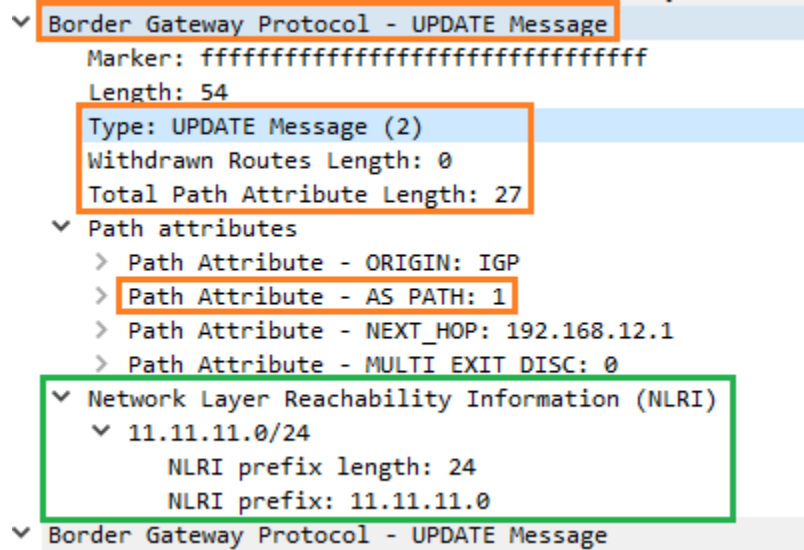
Wireshark capture of an open message between R1 and R2.

The image shows a Wireshark packet capture of an OPEN BGP message. The packet list on the left shows three packets: an Internet Protocol Version 4 packet, a Transmission Control Protocol packet, and a Border Gateway Protocol - OPEN Message packet. The selected packet is the BGP OPEN message. The packet details pane shows the following fields:

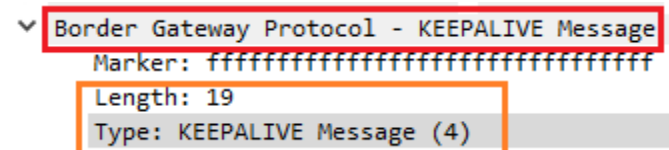
- Marker: ffffffffffffffffffffffffffffffffff
- Length: 58
- Type: OPEN Message (1)
- Version: 4
- My AS: 2
- Hold Time: 180
- BGP Identifier: 22.22.22.22
- Optional Parameters Length: 29

The packet bytes pane shows the raw data of the message.

Wireshark capture of Update message between R1 and R2.



Wireshark capture of Keepalive message between R1 and R2.



Change BGP AS Number

```
R2(config-router)#no router bgp 2
R2(config)#router bgp 22
R2(config-router)#neighbor 192.168.12.1 remote-as 1
```

